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11. Appendix

11.1. Alternative Methods for Scale Membership Probing

Instead of summing up the weakest and strongest adjectives on each half-scale as the scale vector, and ranking the cosine similarity between each adjective and scale vectors for their scale memberships, we also experimented with computing the cosine similarity between each adjective and all adjectives on different scales, and averaging the similarity (we skip the assessed adjective itself when computing similarity with the scale it belongs to). We report results for the best-performing layers below. We observe that using the end adjectives provides better results than all adjectives (i.e. the method in the main content).

Models	DM	CD	WK
fast-text	0.748	0.445	0.958
BERT-b	0.660 $\pm$ 0.009	0.461 $\pm$ 0.020	0.941 $\pm$ 0.012
BERT-l	0.707 $\pm$ 0.013	0.481 $\pm$ 0.014	0.938 $\pm$ 0.013
RoBERTa-b	0.342 $\pm$ 0.021	0.203 $\pm$ 0.015	0.612 $\pm$ 0.043
RoBERTa-l	0.535 $\pm$ 0.016	0.326 $\pm$ 0.016	0.856 $\pm$ 0.019

Table 8: Direct scale membership probing results by comparing averaged cosine similarities between the target adjective to all adjective members on different scales (in comparison to calculating the cosine similarity between the target adjective and the scale vector in the main content). Subscripted numbers are standard deviation in ten runs. The best results per dataset are in bold.

Next, we show some alternatives to mean pooling, the pooling method used in the main content. In contextualized representation computation, we used mean pooling when words are segmented. Instead of mean pooling, we also tried max/min pooling. We report the results below (we use end adjectives in this experiment). Generally, mean pooling performs better.

Models	DM	CD	WK
BERT-b	0.815 $\pm$ 0.009/0.807 $\pm$ 0.006	0.783 $\pm$ 0.011/0.775 $\pm$ 0.010	0.997 $\pm$ 0.004/0.996 $\pm$ 0.004
BERT-l	0.835 $\pm$ 0.011/0.832 $\pm$ 0.011	0.787 $\pm$ 0.010/0.789 $\pm$ 0.009	0.998 $\pm$ 0.003/0.998 $\pm$ 0.003
RoBERTa-b	0.642 $\pm$ 0.009/0.641 $\pm$ 0.008	0.682 $\pm$ 0.007/0.684 $\pm$ 0.006	0.881 $\pm$ 0.023/0.881 $\pm$ 0.016
RoBERTa-l	0.727 $\pm$ 0.010/0.723 $\pm$ 0.006	0.721 $\pm$ 0.007/0.720 $\pm$ 0.008	0.954 $\pm$ 0.011/0.950 $\pm$ 0.014

Table 9: Direct scale membership probing results using min/max pooling (left/right). Subscripted numbers are standard deviation in ten runs.

11.2. Full Table of Scalar Intensity Ranking Results

Model	<i>Dvec</i>	Method	DM			CD			WK		
			P-ACC	τ	ρ	P-ACC	τ	ρ	P-ACC	τ	ρ
BERT-base	DM	G&A	-	-	-	<u>0.735</u>	<u>0.668</u>	<u>0.745</u>	0.902	0.803	0.820
		Ours	-	-	-	0.706	0.604	0.681	0.967	0.934	0.951
	CD	G&A	<u>0.646</u>	<u>0.431</u>	0.509	-	-	-	0.852	0.705	0.756
		Ours	0.639	0.417	<u>0.522</u>	-	-	-	<u>0.902</u>	<u>0.803</u>	<u>0.883</u>
	WK	G&A	<u>0.584</u>	<u>0.303</u>	0.317	<u>0.704</u>	<u>0.602</u>	0.685	-	-	-
		Ours	<u>0.584</u>	0.295	<u>0.360</u>	0.694	0.581	0.659	-	-	-
BERT-large	DM	G&A	-	-	-	<u>0.731</u>	<u>0.663</u>	<u>0.717</u>	0.918	0.836	0.815
		Ours	-	-	-	0.727	0.650	0.703	0.902	0.803	<u>0.873</u>
	CD	G&A	0.695	0.531	0.623	-	-	-	0.918	0.836	0.868
		Ours	0.673	0.488	0.606	-	-	-	<u>0.918</u>	<u>0.836</u>	<u>0.900</u>
	WK	G&A	0.613	0.362	0.372	<u>0.707</u>	<u>0.605</u>	0.649	-	-	-
		Ours	<u>0.628</u>	<u>0.391</u>	<u>0.445</u>	0.706	<u>0.605</u>	<u>0.685</u>	-	-	-
RoBERTa-base	DM	G&A	-	-	-	0.645	0.485	0.536	0.820	0.539	0.727
		Ours	-	-	-	<u>0.748</u>	<u>0.698</u>	<u>0.759</u>	0.934	<u>0.869</u>	<u>0.929</u>
	CD	G&A	0.540	0.207	0.222	-	-	-	0.820	0.640	0.710
		Ours	<u>0.648</u>	<u>0.431</u>	0.558	-	-	-	<u>0.918</u>	<u>0.836</u>	<u>0.893</u>
	WK	G&A	0.557	0.233	0.253	0.599	0.377	0.450	-	-	-
		Ours	<u>0.597</u>	<u>0.326</u>	<u>0.469</u>	<u>0.661</u>	<u>0.500</u>	<u>0.601</u>	-	-	-
RoBERTa-large	DM	G&A	-	-	-	0.682	0.561	0.620	0.836	0.672	0.807
		Ours	-	-	-	0.752	0.702	0.783	0.934	<u>0.869</u>	<u>0.912</u>
	CD	G&A	0.595	0.323	0.378	-	-	-	0.820	0.639	0.690
		Ours	<u>0.664</u>	<u>0.465</u>	<u>0.612</u>	-	-	-	<u>0.902</u>	<u>0.803</u>	<u>0.880</u>
	WK	G&A	0.558	0.241	0.261	0.645	0.478	0.571	-	-	-
		Ours	<u>0.642</u>	<u>0.417</u>	<u>0.544</u>	<u>0.685</u>	<u>0.558</u>	<u>0.673</u>	-	-	-

Table 10: Full evaluation results of applying our methods to different models with additional metrics Kendall’s τ and Spearman’s ρ . The best results across models and methods are marked in bold. Best results using the same model and *Dvec* resource are underlined.

11.3. Construction templates and examples for scale membership indirect probing

Index	Template	Example for adjs on the same scale	Example for adjs on different scales
1	ADJ_{weak} or even ADJ_{strong}	<i>warm</i> or even <i>hot</i>	<i>#warm</i> or even <i>tasty</i>
2	ADJ_{weak} if not ADJ_{strong}	<i>warm</i> , if not <i>hot</i>	<i>#warm</i> if not <i>tasty</i>
3	$ADJ_{weak},$ or even ADJ_{strong}	<i>warm</i> , or even <i>hot</i>	<i>#warm</i> or even <i>tasty</i>
4	$ADJ_{weak},$ if not ADJ_{strong}	<i>warm</i> , if not <i>hot</i>	<i>#warm</i> , if not <i>tasty</i>

Table 11: Construction templates with which adjectives on the same scale are more likely to appear than those on different scales. # means pragmatically bad expressions.